



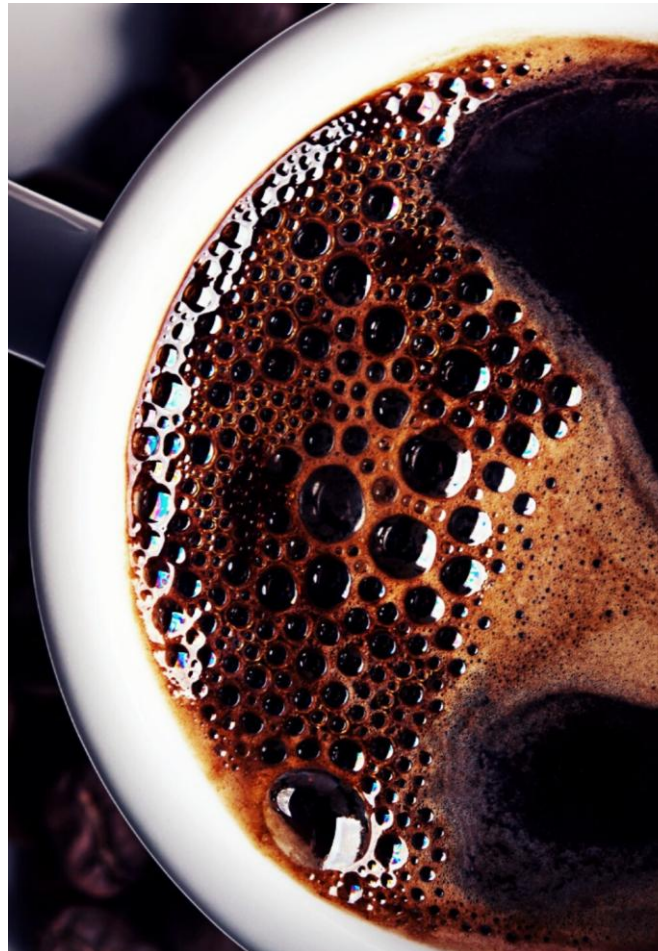
Predicting Global Coffee Quality Using Machine Learning

CQI Dataset Analysis

Lindsey Bowen, Spring 2026

Problem Definition

Aroma	Flavor	Aftertaste	Acidity	Body	Balance	Uniformity	Clean.Cup	Sweetness	Cupper.Points	Total.Cup.
8.67	8.83	8.67	8.75	8.5	8.42	10	10	10	8.75	
8.75	8.67	8.5	8.58	8.4	8.42	10	10	10	8.58	
8.42	8.5	8.42	8.42	8.3	8.42	10	10	10	9.25	
8.17	8.58	8.42	8.42	8.5	8.25	10	10	10	8.67	
8.25	8.5	8.25	8.5	8.4	8.33	10	10	10	8.58	
8.58	8.42	8.42	8.5	8.3	8.33	10	10	10	8.33	
8.42	8.5	8.33	8.5	8.3	8.25	10	10	10	8.5	
8.25	8.33	8.5	8.42	8.3	8.5	10	10	9.33	9	
8.67	8.67	8.58	8.42	8.3	8.42	9.33	10	9.33	8.67	
8.08	8.58	8.5	8.5	7.7	8.42	10	10	10	8.5	
8.17	8.67	8.25	8.5	7.8	8.17	10	10	10	8.58	
8.25	8.42	8.17	8.33	8.1	8.17	10	10	10	8.5	
8.08	8.67	8.33	8.42	8	8.08	10	10	10	8.33	
8.33	8.42	8.08	8.25	8.3	8	10	10	10	8.58	
8.25	8.33	8.5	8.25	8.6	8.75	9.33	10	9.33	8.5	
8	8.5	8.58	8.17	8.2	8	10	10	10	8.17	
8.33	8.25	7.83	7.75	8.5	8.42	10	10	10	8.33	
8.17	8.33	8.25	8.33	8.4	8.33	9.33	10	9.33	8.83	
8.42	8.25	8.08	8.17	7.9	8	10	10	10	8.42	
8.17	8.17	8	8.17	8.1	8.33	10	10	10	8.33	
8	8.25	8.08	8.5	8.3	8	10	10	10	8.17	
8.08	8.25	8	8.17	8	8.33	10	10	10	8.33	
8.17	8.25	8.17	8	7.8	8.17	10	10	10	8.58	
8.25	8.33	8.17	8.17	7.8	8.17	10	10	10	8.17	
8.42	8.17	7.92	8.17	8.3	8	10	10	10	8.08	
8.5	8.5	8	8	8	8	10	10	10	7.92	
7.83	8.25	8.08	8.17	8.2	8.17	10	10	10	8.25	
8.42	8.17	8.17	8.17	7.8	7.92	10	10	10	8.17	
8.17	8.08	8.08	8	8.1	8	10	10	10	8.25	
8	8	8	8.25	8	8.17	10	10	10	8.17	
7.92	8.25	8	8.33	8	8.08	10	10	10	8	
8.42	8.17	8.17	8	7.6	8	10	10	10	8.17	
8.5	8.17	8	7.75	8	8	10	10	10	8	
8.17	7.83	8	8.08	7.8	8	10	10	10	8.42	
8	8.08	7.92	8	8.1	8.08	10	10	10	8.08	
8.08	8	8	8.25	7.9	7.92	10	10	10	8.08	
8.08	8.17	8	7.92	7.9	7.83	10	10	10	8.33	
8.25	8.42	8.08	7.75	7.7	7.83	10	10	10	8.25	
7.92	7.75	8.08	8.08	8.1	8.17	10	10	10	8.17	
7.67	8.17	8	8	8.2	8.08	10	10	10	8.08	
7.83	8.33	7.83	8.25	7.6	7.92	10	10	10	8.42	
8.17	7.92	7.92	7.75	8.3	8	10	10	10	8.08	
8.42	7.92	8	7.75	7.9	8	10	10	10	8.17	
8.08	8.17	7.75	8.08	7.8	7.83	10	10	10	8.42	
8.08	8.08	8	8	7.8	8.08	10	10	10	8	
8.33	8	8	8	7.8	8	10	10	10	8	
8.08	8.08	7.83	7.92	8.1	7.92	10	10	10	8.08	



Coffee quality is subjective, but experts use standardized “cupping scores”, which depend on sensory traits like aroma, flavor, and balance

Today’s Question: Can we predict coffee quality using production data alone?

Why This Matters

Coffee quality is an important factor in price, consumer perception, and market value. Predicting quality could help improve crops and assist buyers in sourcing high-quality coffee.

Did you know?

Finland has the highest per capita of coffee consumption at 12kg per person annually.



Methodology and Model Choice



Methods Chosen were Random Forest and XGBoost, which is high-performing.



Two different models were used, Regression and Classification models.



In the dataset, two csv's are included, Arabica and Robusta. Only Arabica was used for clarity.

Research Questions: What are we trying to find?



Can we predict exact coffee scores?



Can we identify high-quality coffee?



Which features influence quality the most?



Do different models agree with each other?

The Data

The dataset “Coffee Quality database from Coffee Quality Institute was used, which has 44 columns, such as altitude, country of origin, region, processing methods, etc.

The main columns that contribute to the cupping score are Aroma, Flavor, Aftertaste, Body, Balance, and Uniformity, all traits that seasoned coffee tasters look for when cupping coffee.



<https://www.kaggle.com/datasets/volpatto/coffee-quality-database-from-cqi?resource=download>

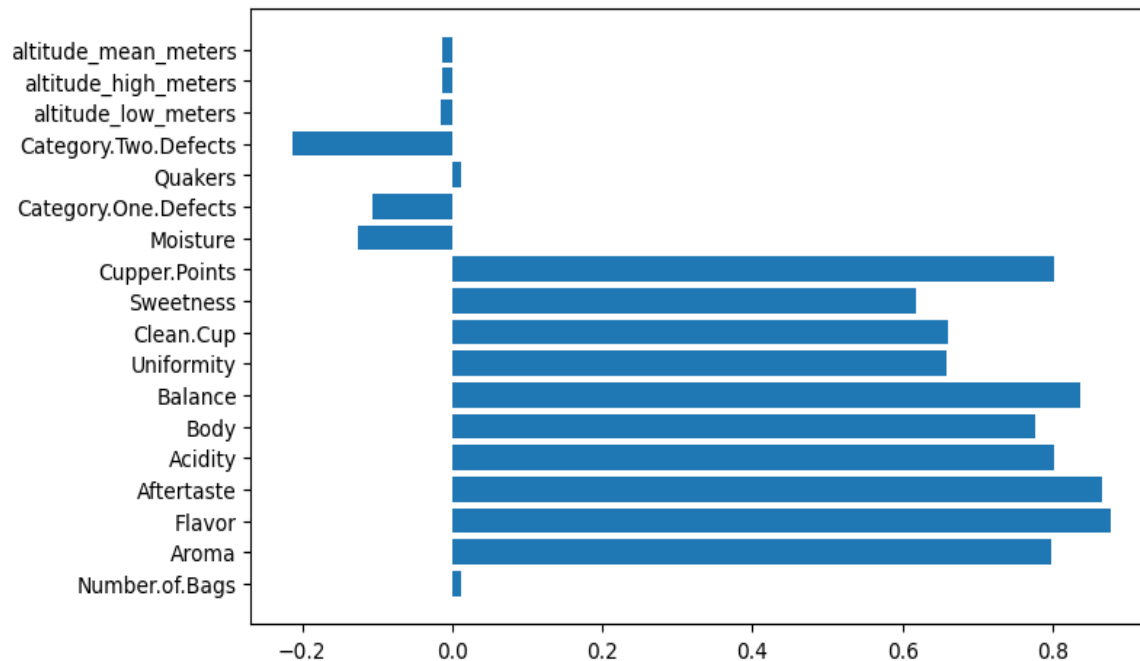
Coffee Quality database from CQI

A database scrapped from Coffee Quality Institute

Statistical Baseline

In this dataset, there are 1310 samples with a mean cupping score of about 82.18.

Visualization of Correlated Features



Statistical Baseline of Total Cupping Points

	Total.Cup.Points
count	1310.00
mean	82.18
std	2.69
min	59.83
max	90.58
range	59.83 – 90.58



XGBoost: Regression and Classification

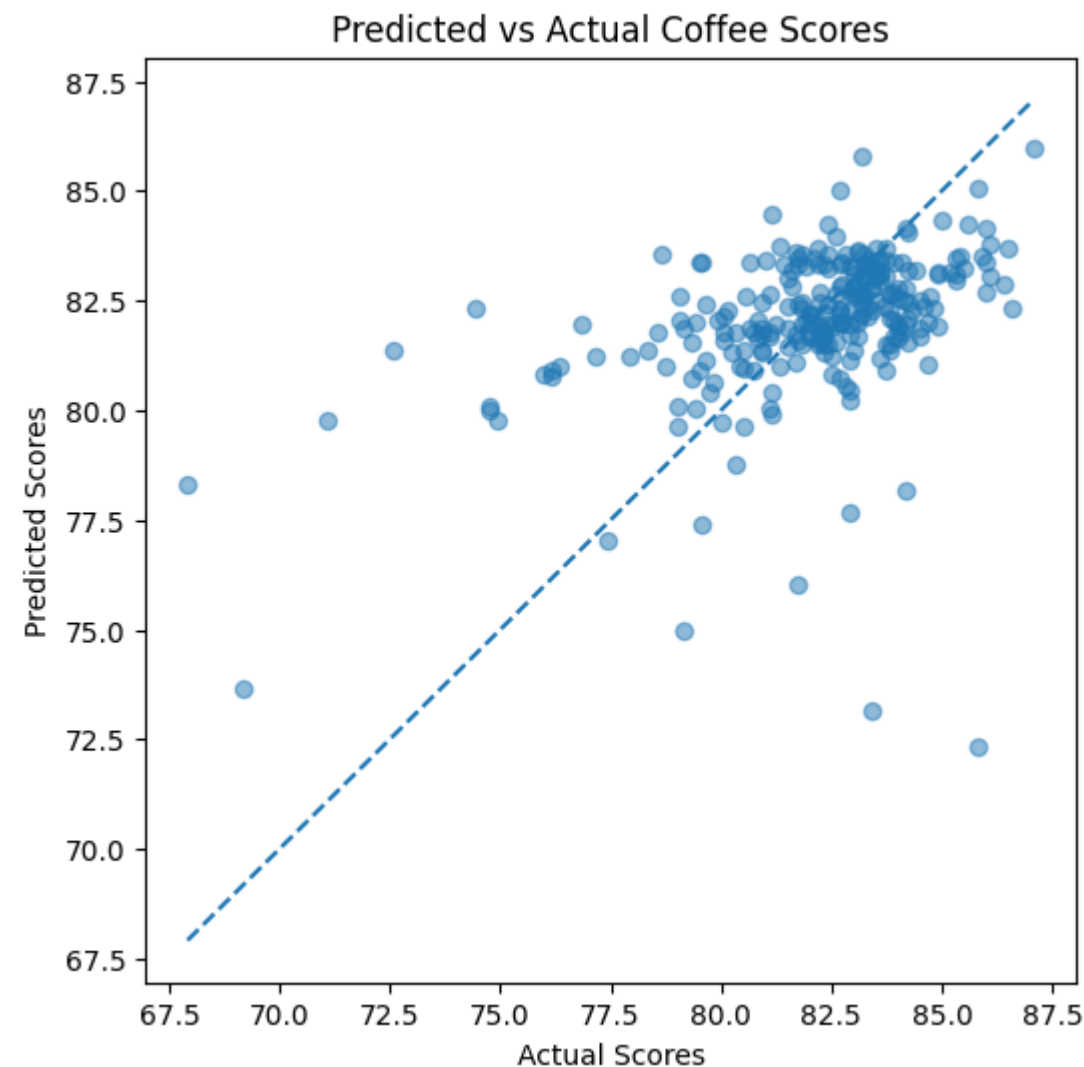
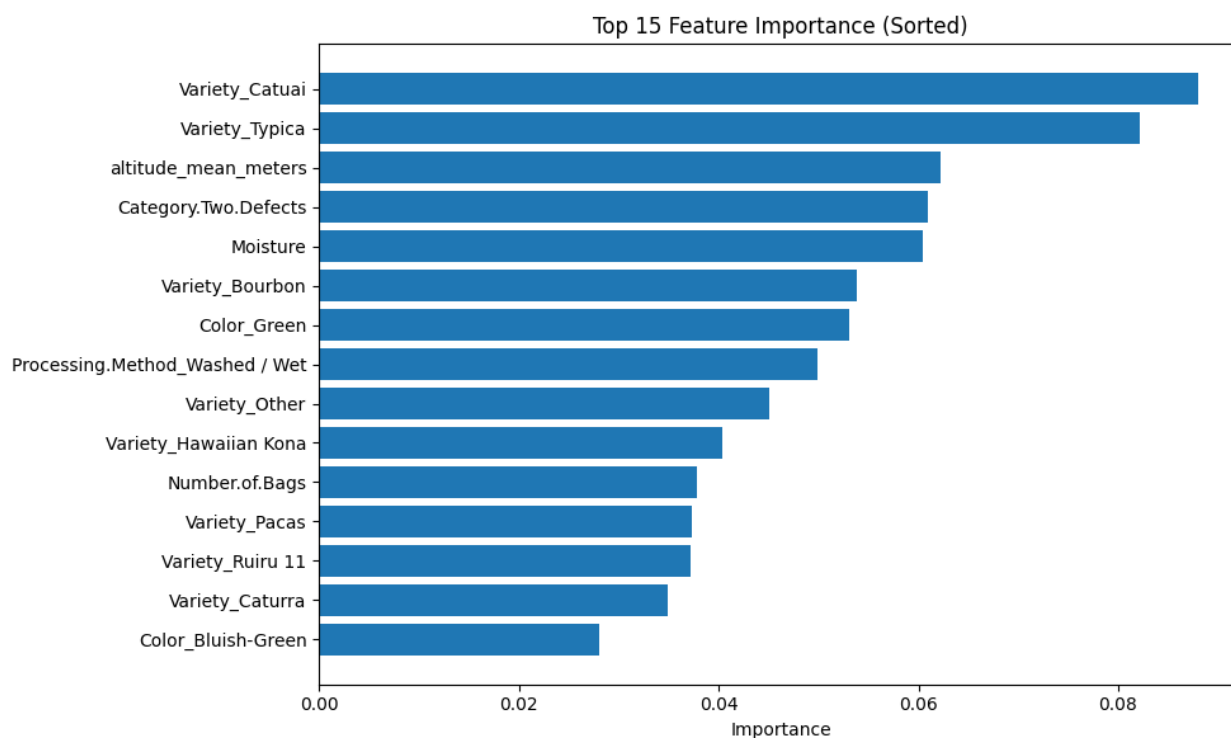
XGBoost: Regression Model

Evaluation Metrics:

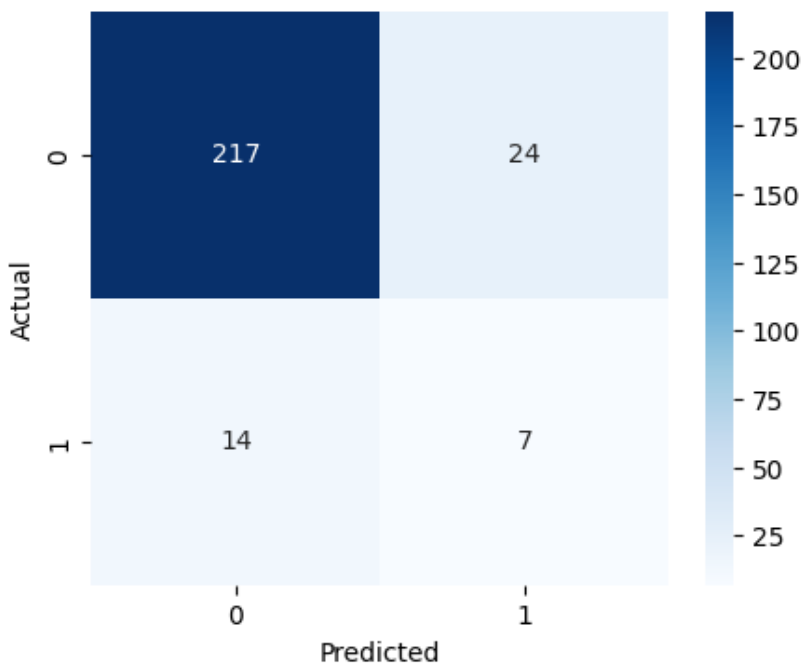
Root Mean Squared Error: 2.27279

R^2 : 0.14618214105257188

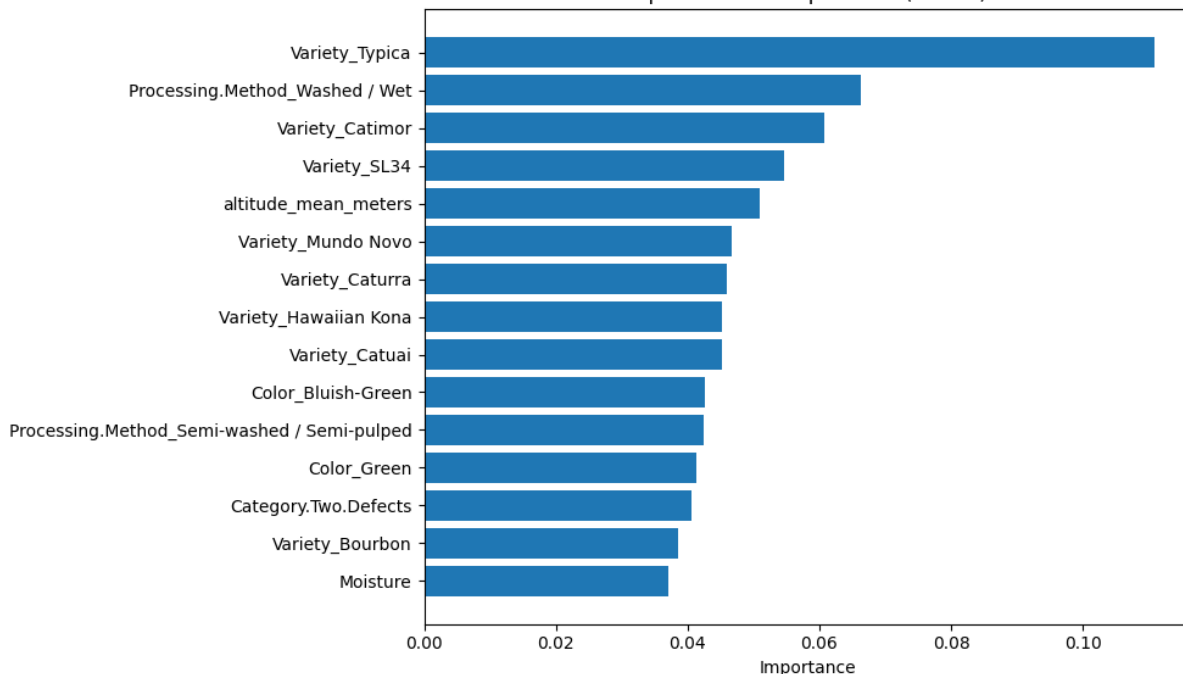
MAE: 1.6772725368820074



Confusion Matrix



Top 15 Feature Importance (Sorted)



XGBoost: Classification Model

- A baseline model was built to predict “high-quality” coffee versus “low-quality”.
- Since only about 8% of samples were high quality, class weighting was used to account for class imbalance and improve the detection of those rarer cases.

Evaluation Metrics:

Accuracy: 0.85496

Precision: 0.23

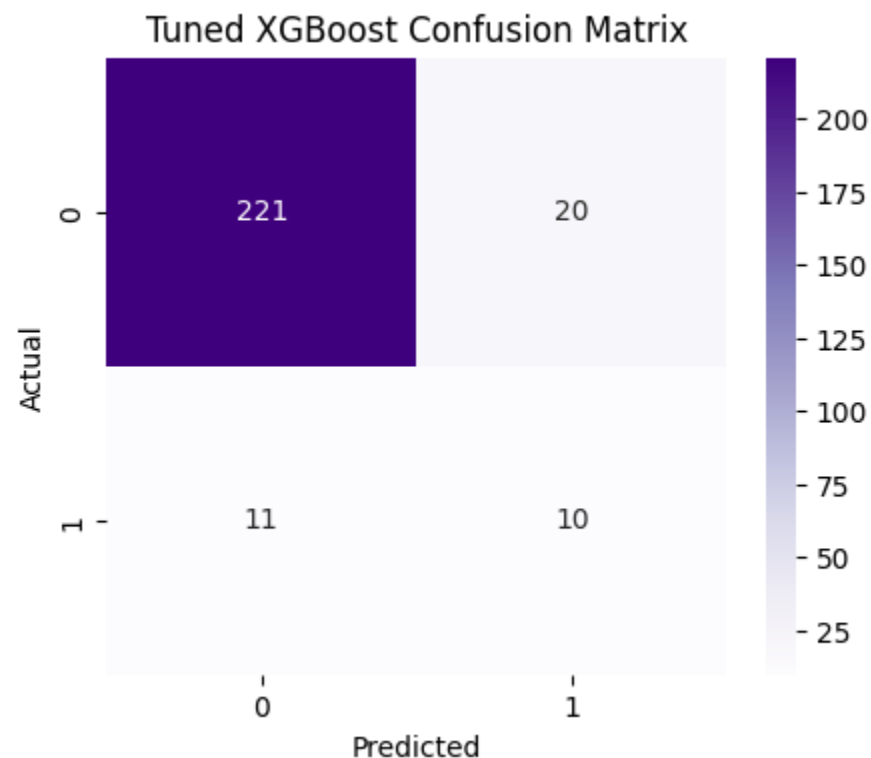
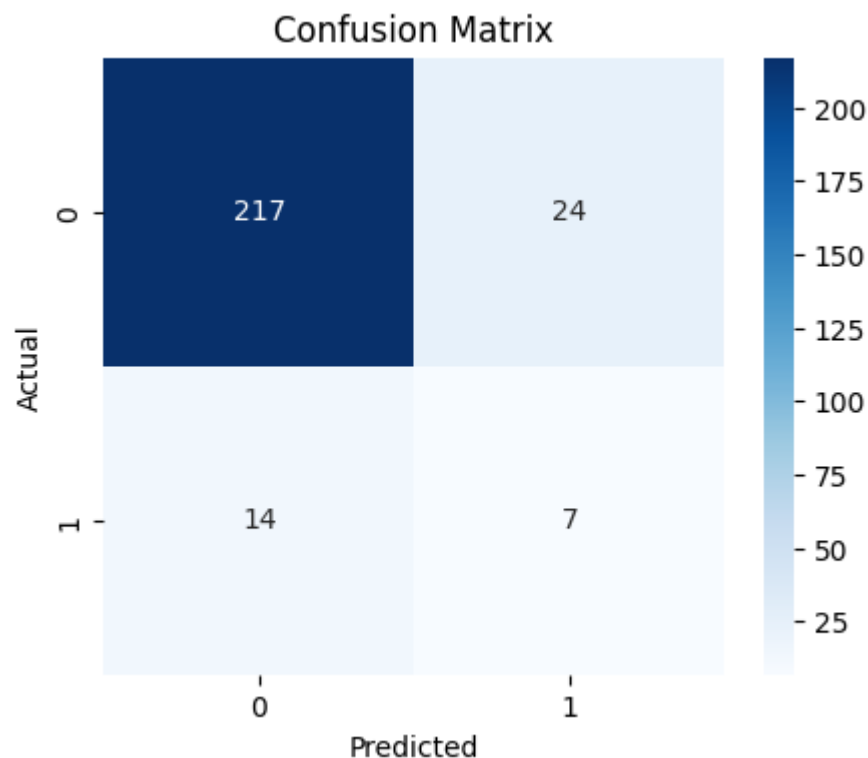
Recall: 0.33

F1 Score: 0.27

Hyperparameter Tuning For XGBoost Classifier

Tuned parameters: n_estimators,,
max_depth, learning_rate

	Accuracy	Precision	Recall	F1-Score
Base	0.844962	0.22806	0.33333	0.269231
Tuned	0.881679	0.33333	0.47619	0.392157



Random Forests: Regression and Classification



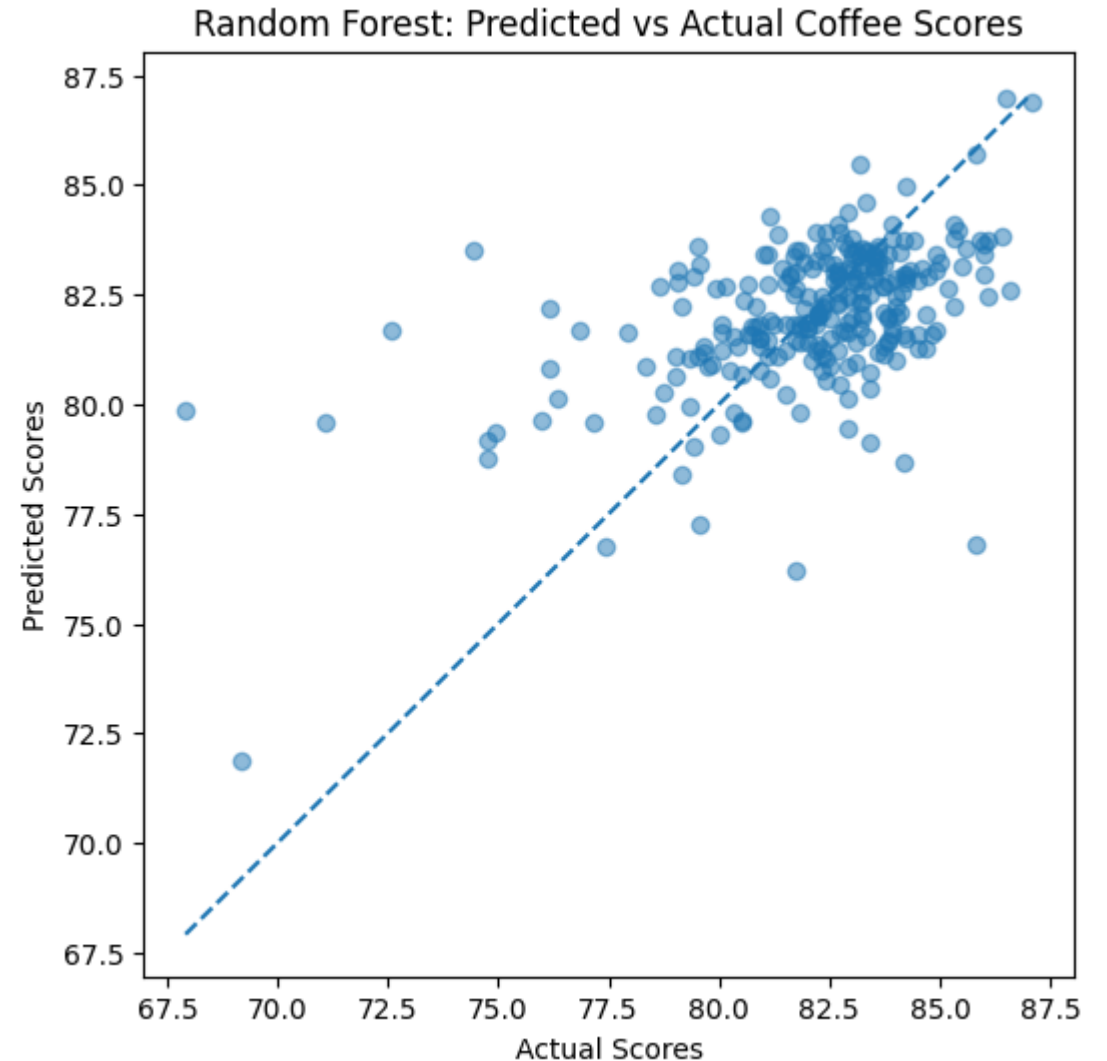
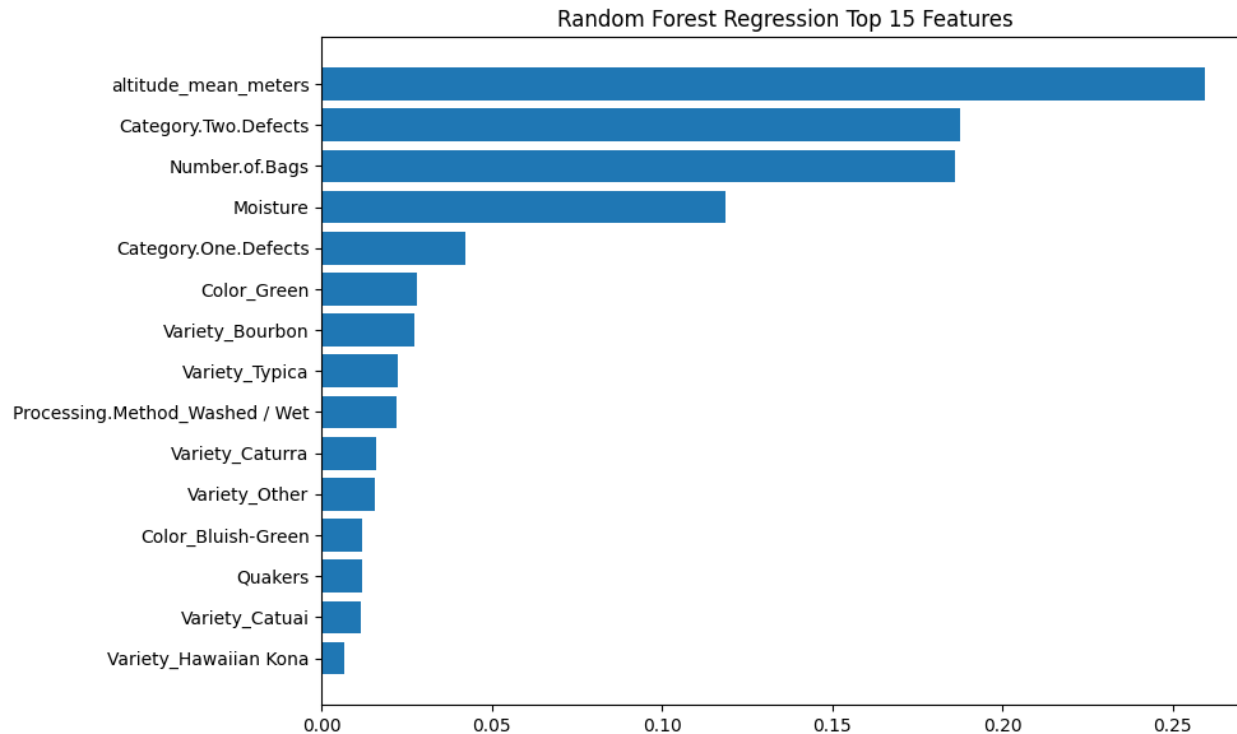
Random Forest: Regression Model

Evaluation Metrics:

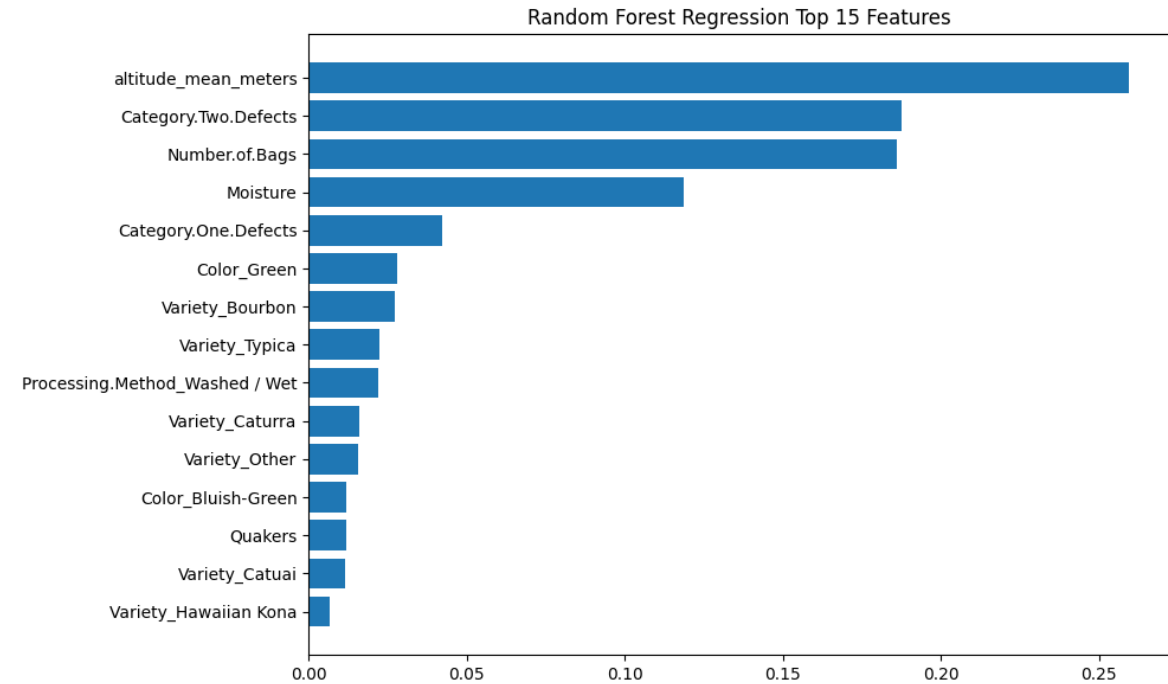
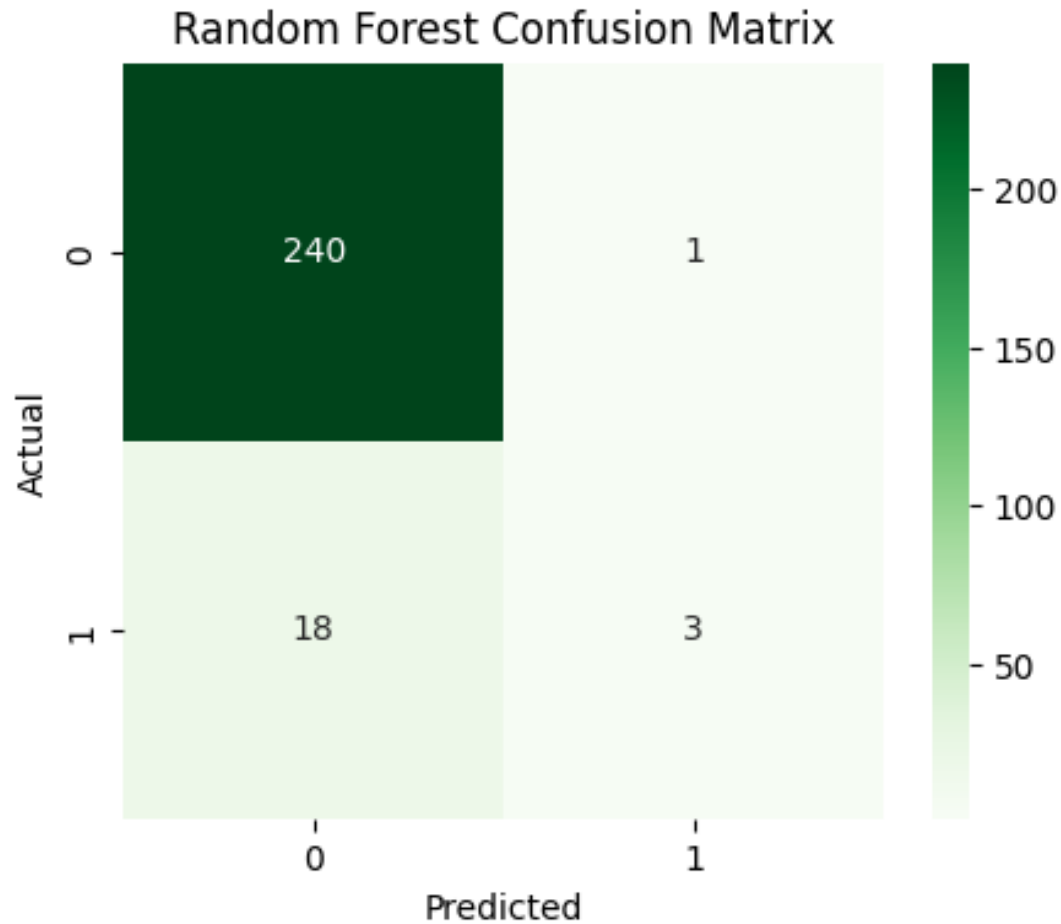
Root Mean Squared Error: 2.2825

R^2 : 0.26442999262152056

MAE: 1.602423719859442



Random Forest: Classification Model



Evaluation Metrics:

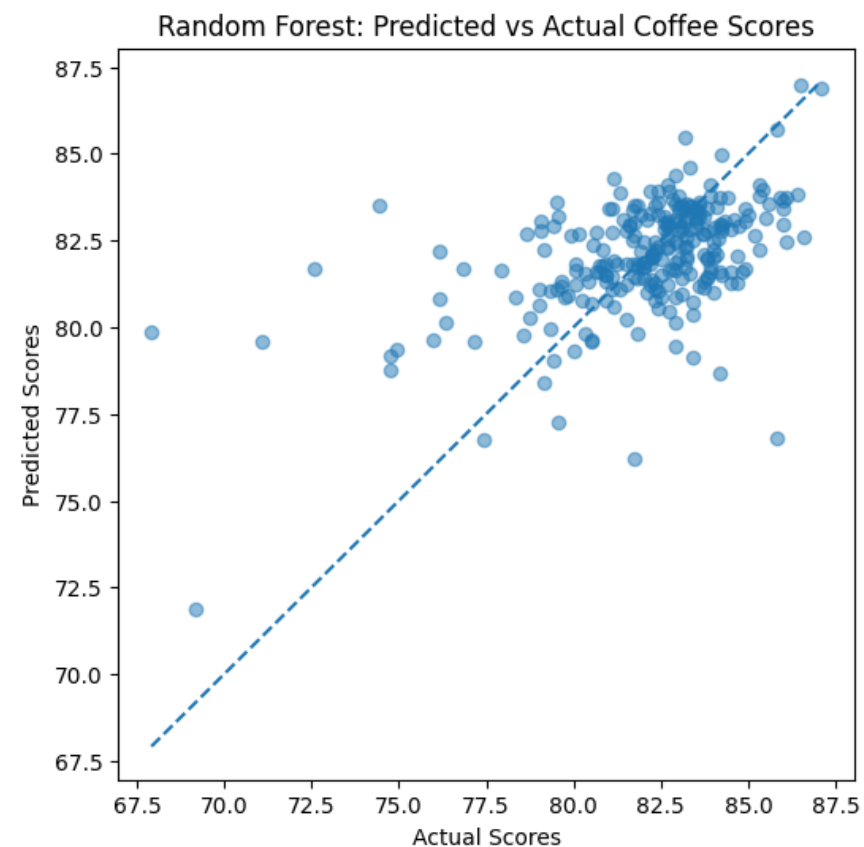
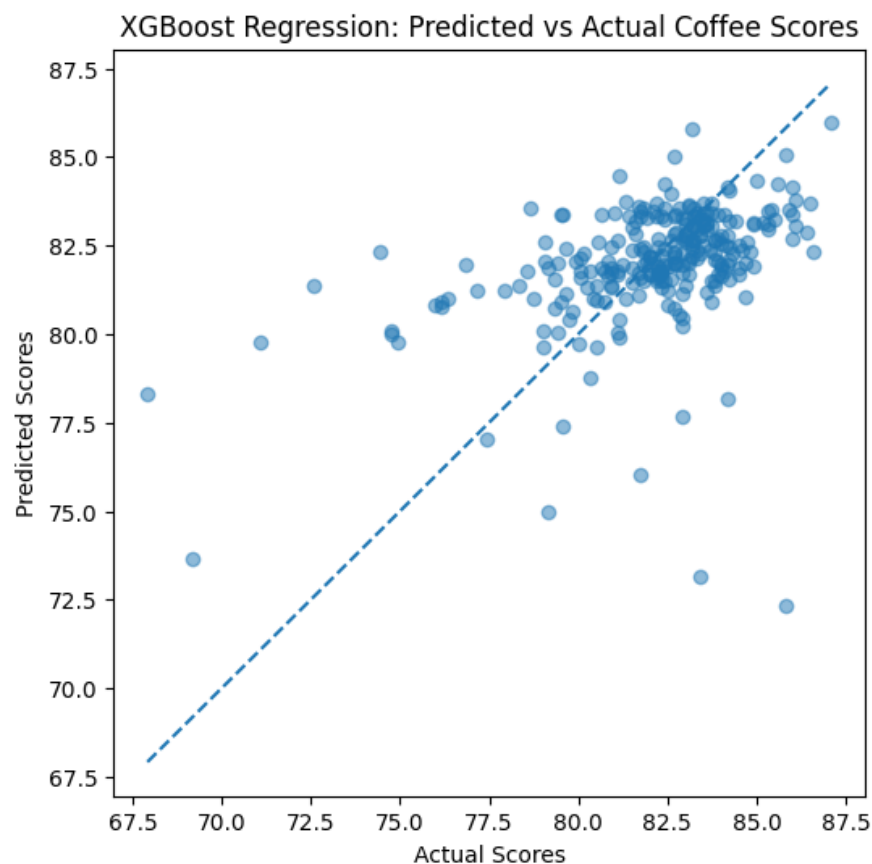
Accuracy: 0.92748

Precision: 0.75

Recall: 0.14

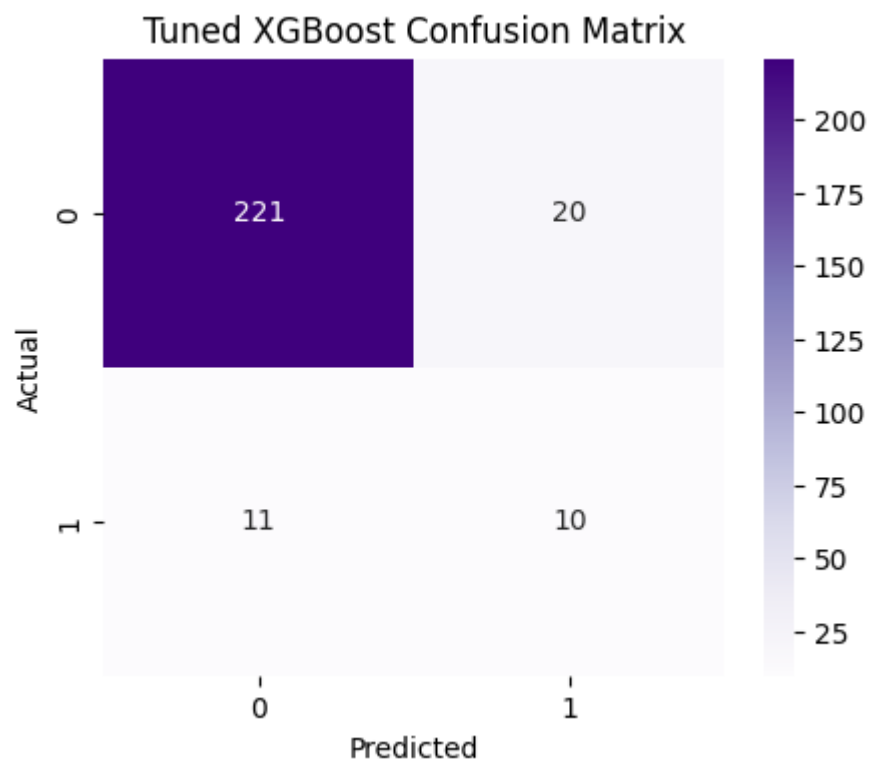
Comparison of XGBoost and RF Regression Models

Model	RMSE	R ²
XGBoost	2.27279	0.14618
RF	2.2825	0.26443

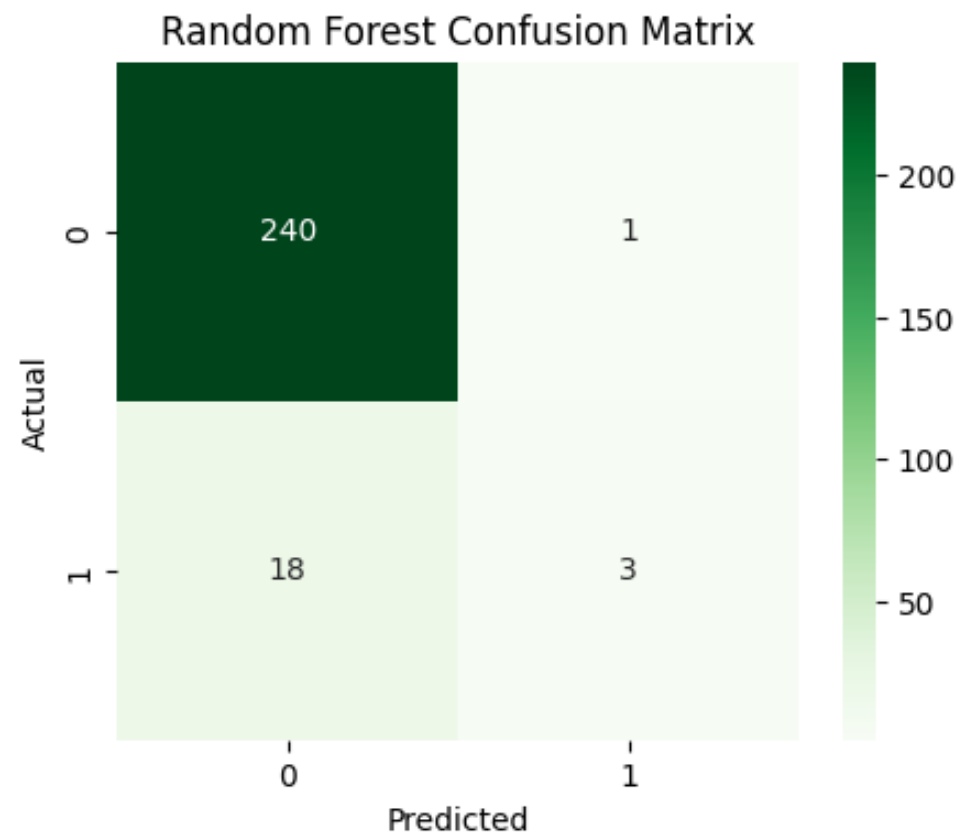


Comparison of XGBoost and RF Classification Models

Accuracy: 0.881679
Precision: 0.33
Recall: 0.48
F1-Score: 0.39



Accuracy: 0.92748
Precision: 0.75
Recall: 0.14
F1-Score: 0.24



Errors and Limitations

- Dataset is highly imbalanced (~8% high-quality)
- A lot of production features show weak correlation with quality
- Some features (e.g., Number of Bags) improve prediction but may reflect dataset-specific biases rather than causal relationships.





Conclusion

- Coffee quality is difficult to predict using production features alone
- Regression models showed limited predictive power (low R^2)
- Classification improved with hyperparameter tuning
- Tree based models captured nonlinear relationships better than linear models

Final Takeaway: Model performance is limited more by data quality than model choice.